



CyberPatriot Mint21

Practice Image Answer Key



Welcome to the CyberPatriot Practice Round! This image will provide you with information on how to solve common vulnerabilities on a Mint 21 operating system. In doing so, it will help you on your way as you build your cybersecurity skills.

The vulnerabilities in this image are some of the most basic ones found during a CyberPatriot competition. Even if you do very well with these vulnerabilities, you will experience greater difficulty as the season progresses. The README file on the desktop in this image may be more detailed than those you see during the competition. You will have to use your own knowledge, not just the hints in this file, to achieve a high score during the actual competition.

Below are the answers to the problems that are being scored in this image. Each one includes information on how the problem was found (if applicable), how it was solved, and why it is important from a cybersecurity standpoint.

It is also possible to lose points during the competition. Simple penalties that may arise are noted below the answers. There are many ways to solve some of the problems below. This answer key just shows one method in each case.

Answers

1) Forensics Question 1 Correct: 8 pts.

When you open an image, please read all the "Forensics Questions" thoroughly before modifying the image as you may change something that prevents you from answering the question correctly. There is a file on the Desktop here named "Forensics Question 1".

Press Control + Alt + T to open a Terminal. Enter the command **sudo updatedb** and enter your password (found in the ReadMe). Next, enter the command **locate *.mp3** to view all mp3 files on the system.

```
benjamin@mint: ~  
File Edit View Search Terminal Help  
benjamin@mint:~$ sudo updatedb  
[sudo] password for benjamin:  
benjamin@mint:~$ locate *.mp3  
/home/benjamin/Music/01-01- Ocean Motion.mp3  
/home/benjamin/Music/01-02- Fall Festival.mp3  
/home/benjamin/Music/01-03- Flight of the Humble Bee.mp3  
/home/benjamin/Music/01-04- Tinker Bells Dream.mp3  
/home/benjamin/Music/01-05- Vertigo Delight.mp3  
/home/benjamin/Music/01-06- Much Ado About Something.mp3  
/home/benjamin/Music/01-07- Last Thoughts.mp3  
/usr/share/javascript/mathjax/unpacked/extensions/ally/invalid_keypress.mp3  
benjamin@mint:~$
```

The file path that contains all of the files is **/home/benjamin/Music**.

2) Removed unauthorized user **tgianopolous**: 5 pts.

One of the first things you should do when starting an image during a competition is check the README file on the desktop. The authorized administrators and users listed in the README are the only users that should exist on the system (aside from legitimate built-in system accounts and those used for services). All unauthorized user accounts should be removed.

Open the MATE Menu  in the lower left corner of the desktop. Click on Administration and Users and Groups. Scroll through the list of users and find **tgianopolous**. Click on the user and select Delete. You may need to enter your password to delete users. After you enter your password, click Authenticate → Delete Files.

3) Removed unauthorized user **jqelling**: 5 pts.

Open the MATE Menu  in the lower left corner of the desktop. Click on Administration and Users and Groups. Scroll through the list of users and find **jqelling**. Click on the user and select Delete. You may need to enter your password to delete users. After you enter your password, click Authenticate → Delete Files.

4) User **dscott** is not an administrator: 5 pts.

Open the MATE Menu  in the lower left corner of the desktop. Click on Administration and Users and Groups. In Users Settings, scroll through the list of users and click on **jqelling**. Next to Account Type:, click on Change.. → Desktop User → OK. You may need to enter your password to change the account type. Enter your password in the next window and click Authenticate.

5) Changed insecure password for user **llitt**: 5 pts.

In the ReadMe, this administrator's password only has 10 characters. Open the MATE Menu  in the lower left corner of the desktop. Click on Administration and Users and Groups. In Users Settings, scroll through the list of users and click on **llitt**. Next to Password, click on Change... Enter your password and click Authenticate. In the Change User Password window, make sure Set password by hand is selected. Enter a secure password with at least 12 characters in the New password: field. Enter the same password in the Confirmation: field below and click OK.

6) Created new administrator account for **edarby**: 6 pts.

Press Control + Alt + T to open a Terminal. Enter the following commands:

```
sudo adduser edarby
```

```
sudo usermod -aG sudo edarby
```

Enter your password if you are prompted after running each command.

7) User **edarby** must change password at next login: 6 pts.

Press Control + Alt + T to open a Terminal. Enter the following commands:

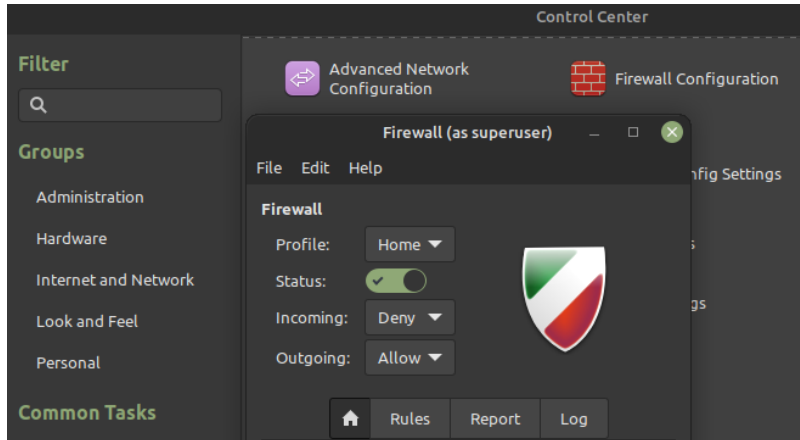
```
sudo passwd -e edarby
```

Enter your password if you are prompted after running the command.

8) Uncomplicated Firewall (UFW) protection has been enabled: 6 pts.

Enabling a host-based firewall is very important to system security. The README tells you that the only company approved firewall is UFW. You can check the status of UFW by typing **sudo ufw status**. If prompted by sudo for a password, type your current user's password. The password for your current user account can be found in the README.

Inside the MATE Menu , on the left side of the panel, click on **Control Center**. Inside the **Control Center** window, click on Firewall Configuration. If prompted type the password of your current user account. The password for your current user account can be found in the README. Click the toggle button next to Status to enable the firewall. Alternatively, you can type **sudo ufw enable** on the command line.



9) FTP service has been disabled or removed: 6 pts.

Press Control + Alt + T to open a Terminal. Enter the following commands:

```
sudo systemctl stop vsftpd
```

```
sudo apt purge vsftpd
```

In the output, you will be prompted Do you want to continue? [Y/n]

Type **y** and press **Enter**.

10) The system refreshes the list of updates automatically: 5 pts.

Automatically installing updates is good cybersecurity practice.

Inside the MATE Menu , on the left side of the panel, click on Control Center. Inside the Control Center window, click on **Update Manager**. Select Edit → Preferences and click the Automation tab inside the Preferences window. In the Automation tab, select Apply updates automatically. If prompted type the password of your current user account. The password for your current user account can be found in the README.

11) Install updates from important security updates: 5 pts.

In **Update Manager**, make sure you have set in Preferences to Apply updates automatically. Instructions on how to do this are listed above under **The system refreshes the list of updates automatically**.

Close out of Preferences and click on Refresh in the **Update Manager**. Then click on Install Updates.

Alternatively, you can enter the following commands in the Terminal:

sudo apt update

sudo apt full-upgrade -y

Apt may ask you some questions that require an answer, such as asking if configuration files should be updated or kept for openssh-server. Keeping the local version is usually the safer choice, and changing a configuration file may break services.

12) Systemd, Chromium, OpenSSH have been updated: 5 pts. each

It is recommended that you update all of the packages on your system at the same time. To do this, run the following commands in the Terminal:

sudo apt update

sudo apt full-upgrade -y

Apt may ask you some questions that require an answer, such as asking if configuration files should be updated or kept for openssh-server. Keeping the local version is usually the safer choice, and changing a configuration file may break services. However, installing the package maintainer's version is sometimes the more secure choice. In this specific case, for openssh-server, it is safe to select install the package maintainer's version but that might not always be the case during a competition.

13) Prohibited MP3 files are removed: 5 pts.

The README specifically states that non-work related media files are prohibited. There are several ways and commands that can be used to find files and file types including locate, find, and file.

In a terminal type `locate '*.mp3'`. In the output of locate you can see the file system location of files with the extension .mp3. The mp3 files under benjamin's Music directory appear to be non-work related. Type **sudo rm /home/benjamin /Music/*.mp3** If prompted by sudo for a password, type your current user's password. The password for your current user account can be found in the README.

14) Prohibited software Wireshark and Zangband removed: 6 pts. each

Removing unauthorized and potentially unwanted programs from a computer is an important cybersecurity principle. Third party software installed on the system should be limited to the software listed in the README, and software required for normal operation of the operating system and services. In this case you can see links to Wireshark and Zangband on the MATE desktop.

On the command line run **sudo apt purge -y wireshark* zangband***

15) SSH root login has been disabled: 6 pts.

OpenSSH Server is listed in the README as a critical service. It's important to research how to secure critical services without breaking the required functionality of the service.

In a terminal run the following commands:

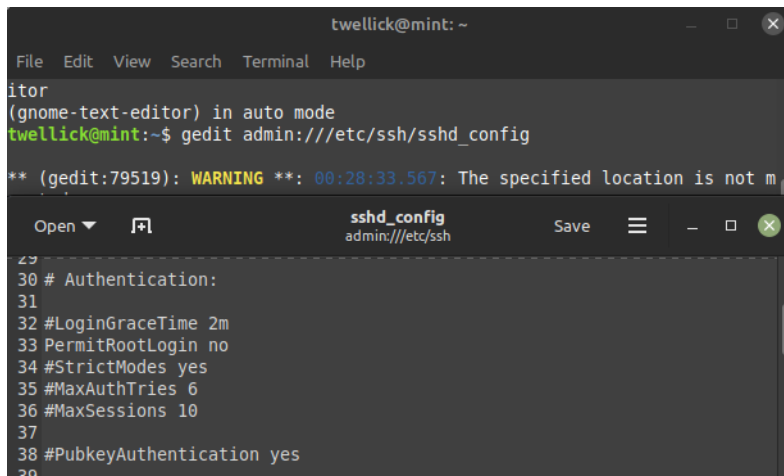
sudo apt update

sudo apt install -y gedit

gedit admin:///etc/ssh/sshd_config

If prompted type the password of the current user account and click Authenticate. The password for the current user account can be found in the README.

Change **PermitRootLogin yes** to **PermitRootLogin no**. Save the file and exit.



The screenshot shows a terminal window titled 'twellick@mint: ~'. The terminal output includes the command 'gedit admin:///etc/ssh/sshd_config' and a warning message: '** (gedit:79519): WARNING **: 00:28:33.567: The specified location is not m'. Below the terminal, a window titled 'sshd_config' is open, showing the configuration file's contents. The configuration includes settings for authentication, login grace time, root login, strict modes, max auth tries, max sessions, and pubkey authentication.

```
twellick@mint: ~
File Edit View Search Terminal Help
itor
(gnome-text-editor) in auto mode
twellick@mint:~$ gedit admin:///etc/ssh/sshd_config
** (gedit:79519): WARNING **: 00:28:33.567: The specified location is not m

Open  sshd_config  Save  -  +  x
admin:///etc/ssh

29
30 # Authentication:
31
32 #LoginGraceTime 2m
33 PermitRootLogin no
34 #StrictModes yes
35 #MaxAuthTries 6
36 #MaxSessions 10
37
38 #PubkeyAuthentication yes
39
```

Penalties

1) **WARNING: VirtualBox is unsupported: -0 pts.**

- Why is this a penalty?

This is a warning to inform the user that VirtualBox is **NOT** supported to run the CyberPatriot 18 competition images. Failure to use the version of VMWare Workstation Pro specified on the CyberPatriot website may result in issues that will not be considered for appeals.

2) **OpenSSH service has been stopped or removed: -5 pts.**

- Why is this a penalty?

The README specifies that the OpenSSH server is a critical service.

3) **Chromium has been removed: -5 pts.**

- Why is this a penalty?

The README specifies that Chromium is the default web browser that must remain on the system.